

Course- BTech
Course Code- CSET101

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Type- Core
Course Name- Computational
Thinking and Programming
Semester- odd
Batch- ALL

Tutorial Assignment: 4
Tutorial title: Practicing Conditional Statements

CO Mapping

Name	CO1	CO2	CO3
Conditional Statements	✓	-	-

CONDITIONAL STATEMENT

if-else: The if statement alone tells us that if a condition is true it will execute a block of statements and if the condition is false it won't. But what if we want to do something else if the condition is false. Here comes the else statement. We can use the else statement with if statement to execute a block of code when the condition is false.

Syntax:

if (condition):

 # Executes this block if

 # condition is true

else:

 # Executes this block if

 # condition is false

Elif: Here, a user can decide among multiple options. The if statements are executed from the top down. As soon as one of the conditions controlling the if is true, the statement associated with that if is executed, and the rest of the ladder is bypassed. If none of the conditions is true, then the final else statement will be executed.

Syntax:

if (condition):

 statement

elif (condition):

 statement

.

.

else:

 statement

nested if-else: A nested if is an if statement that is the target of another if statement. Nested if statements mean an if statement inside another if statement. Yes, Python allows us to nest if statements within if statements. i.e, we can place an if statement inside another if statement.

Syntax:

EXERCISE

Q1: What is the output of following code:

```
tick=0
while(tick<10):
    tick=tick+3
    print("hello")
```

- a) hello
hello
hello
- b) hello hello hello hello
- c) hello
hello
hello
hello
- d) none of these

Q2: What is the output of following code:

```
a='Hello Bennett Students'
i=0
while i <len(a):
    i+=1
    pass
print(i)
```

- a) 22
- b) 20
- c) 21
- d) None of these

Q3: What is the output of the following code:

```
i=0
while i<6:
    i+=1
    print(i)
    break
else:
    print("No break")
```

- a) 1
2
3
4

- b) 1
- 2
- 3
- c) 1
- d) None of these

Q4: Tell the output of python code:

```

if i<j:
    if j<k:
        i=j
    else:
        j=k
else:
    if j>k:
        j=i
    else:
        i=k
print(i,j,k)

```

What is the output if variables i,j and k have the following values?

- a) i=3,j=5,k=7
- b) i=-2,j=-5,k=9
- c) i=8,j=15,k=12
- d) i=13,j=15,k=13

Q5: Tell the output of python code:

```

i=0
a=[5,4,3,2,1]
while i<len(a):
    if a[i]==4:
        i+=1
        continue
    print(a[i])
    i+=1

```

- a) 5
- 4
- 3
- 1
- b) 5
- 3
- 2
- 1
- c) 5
- 4

3

2

d) None of these

Q6: Tell the output of python code when input is num1=123 and num=1234:

```
num1=input("enter any number")
l=len(num1)
if l!=3:
    print("Enter three digit number")
else:
    print("Middle digit is",(int(num1)%100//10))
```

Q7: What is the purpose of else in if elif ladder and whether it is compulsory to end if elif ladder with else statement?

Q8: Tell the output of python code:

```
word=input()
for char in range(len(word)-1,-1,-1):
    print(word[char],end="")
print("\n")
```

Input: Welcome to Bennett

a) ttenneB ot emocleW

b) emocleW ot ttenneB

c) Welcome to Bennett

d) Error in code

Q 9: What is the correct output of following code:

```
numbers=(1,2,3,4,5,6,7,8,9)
count1=0
count2=0
for x in numbers:
    if not x%2:
        count1+=1
    else:
        count2+=1
print(count1)
print(count2)
```

a) (2,4,6,8)

(1,3,5,7,9)

b) (1,3,5,7,9)

(2,4,6,8)

c) 4

5

d) 5

Q 10: What is the correct output of following code:

```
for fizzbuzz in range(16):
    if fizzbuzz % 3 ==0 and fizzbuzz % 5 ==0:
        print("fizzbuzz")
        continue
    elif fizzbuzz % 3 ==0:
        print("fizz")
        continue
    elif fizzbuzz % 5 == 0:
        print("buzz")
        continue
    print(fizzbuzz)
```

Q 11: What is the correct output of the following code:

```
ListA=[1,2,3,4,5]
removed_element=ListA.pop(2)
print(removed_element)
print(ListA)
```

a) 3

b) 2

[1, 3, 4, 5]

c) 4

[1, 2, 3, 5]

d) None of these

Q12: Correct the code to print type of all the items in datalist. Use for loop and rewrite print statement.

```
datalist=[1452,10.23,1+5j,True,'Bennett',(0,3),[5,10],
{"class":'B.tech',"Subject":'Python'}]
```

Q13: Run the code on given input:

a) 330, 300

b) 300,300

c) 215,300

```
a=int(input())
```

```
b=int(input())
```

```
print("A") if a>b else print("") if a==b else print("B")
```

Q14: Run the code on the following input:

a) 0001,0010,0011,0000

b) 0001,0011,0101,1100

```
items=[]
```

```
num=[x for x in input().split(',')]
```

```
for p in num:
    x=int(p,2)
    if not x%5:
        items.append(p)
print(''.join(items))
```

Q15: Write python code using if elif and else ladder to convert month name to a number of days.

Bonus Question:

Write a python program to construct the following pattern, using a nested for loop.

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```